

# Randomness, Probability, and $1/f$ Noise as Frame-Edge Signatures

A Lambda / TuFT / NSAF Formal Note on Sampling, Measure, and Scale-Free Residue

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### Abstract

This note states a frame-theoretic account of randomness, probability, and  $1/f$  noise within the Lambda / TuFT / NSAF vocabulary. The central claim is not that every discrete sampling process produces  $1/f$  noise. Independent coin tosses produce white noise in the raw sequence, and random-walk accumulation produces Brownian-type spectra closer to  $1/f^2$ . The mathematically valid claim is more specific and stronger: when a discrete observational frame samples or reconstructs a richer global process through a scale-free hierarchy of unresolved relaxation or integration timescales, the aggregate fluctuation spectrum has a natural  $1/f$ -type form in an intermediate frequency regime.

Probability supplies the continuum-side measure structure. Random outcomes supply discrete sampled facts. Statistical laws describe how finite or countable sampling relates to the measure structure without collapsing it into any one finite sequence. In Lambda terms, randomness and probability expose a frame edge between individual discrete observations and the global measure object those observations estimate.  $1/f$  noise then appears as a particular spectral residue of this frame edge when the reconstruction process contains no privileged timescale.

### 1. Frame Declaration

The discussion uses two declared descriptive frames.

**Linear / discrete frame  $P_L$ .** This frame contains individual outcomes, finite strings, finite sample paths, empirical frequencies, computable summaries, finite precision records, and arithmetic encodings of observed data.

**Curved / global frame  $P_C$ .** This frame contains probability measures, distribution laws, sigma-algebras, limiting objects, spectral measures, Hilbert-space or phase-space structures, and continuum-level constraints.

A frame transition occurs when an object or process described in  $P_C$  is observed, sampled, encoded, approximated, or reconstructed inside  $P_L$ . The Lambda residue at the seam is the structured information lost, deferred, or externally supplied when the lower frame attempts to close over the higher frame.

This note concerns a specific family of seams:

$$P_C \longrightarrow P_L$$

where a global measure or scale-rich process is sampled through discrete events.

## 2. Probability as a Global Measure Object

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where  $\Omega$  is the sample space,  $\mathcal{F}$  is a sigma-algebra of events, and  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  is a countably additive measure with  $\mathbb{P}(\Omega) = 1$ .

An individual outcome

$$\omega \in \Omega$$

is a discrete realized fact once observed. But the probability law is not identical to any one outcome, nor to any one finite string of outcomes. The law belongs to the global frame. The sample belongs to the discrete frame.

For a fair coin, one may take

$$\Omega = \{0, 1\}^{\mathbb{N}},$$

with the Bernoulli product measure satisfying

$$\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = 0) = \frac{1}{2}.$$

A finite observed string

$$(x_1, x_2, \dots, x_N)$$

is an element of the linear frame. The Bernoulli measure on infinite paths is a global object. The law of large numbers relates these frames by stating, in one standard form, that

$$\frac{1}{N} \sum_{n=1}^N X_n \rightarrow \frac{1}{2}$$

almost surely as  $N \rightarrow \infty$ . The convergence statement uses the measure structure; it is not contained in any finite prefix by itself.

Thus probability already exhibits the Lambda pattern: finite samples can estimate the law, but the law does not reduce to the finite samples. The residue is not an error in probability. It is the structural difference between sampled fact and measure object.

### 3. Randomness as Sample-Measure Non-Collapse

Randomness may be described at several levels.

1. **Classical probabilistic randomness:** outcomes are modeled by a probability measure, and typicality is evaluated by measure-theoretic laws.
2. **Statistical randomness:** finite or infinite sequences pass specified tests for frequency, independence, correlation, or distributional fit.
3. **Algorithmic randomness:** an infinite sequence resists compression or passes effective null-set tests, depending on the chosen formalization.

These are not identical definitions, but they share a frame structure. A discrete sequence is tested against a global law or against a family of effective descriptions. When no finite or admissible description closes over the sequence without remainder, the sequence exhibits randomness relative to that frame.

This permits the following Lambda reading:

Randomness is not mere absence of structure. It is the failure of a declared lower descriptive frame to compress, predict, or finitely close over a sampled path while preserving the global measure constraints under which that path is evaluated.

This statement does not require that randomness be mysterious or non-mathematical. On the contrary, probability theory and algorithmic randomness are rigorous because they define exactly which frame, law, test, or compression class has been declared.

### 4. White Noise, Brownian Noise, and 1/f Noise Must Be Distinguished

A mathematically sound account must distinguish the following cases.

#### 4.1 Independent sampling gives white noise

Let  $X_n$  be independent, identically distributed random variables with finite variance and zero mean. The raw discrete-time process has autocorrelation

$$C(k) = \mathbb{E}[X_n X_{n+k}],$$

with

$$C(k) = 0 \quad \text{for } k \neq 0$$

in the ideal independent case. Its power spectral density is flat, so the raw sequence is white noise. Thus discrete sampling of a probability continuum does not by itself generate 1/f noise.

## 4.2 Random-walk accumulation gives Brownian-type spectra

If

$$Y_N = \sum_{n=1}^N X_n,$$

then  $Y_N$  is an accumulated random walk. Integration of white noise shifts the spectral behavior toward Brownian or red-noise scaling, commonly represented by a  $1/f^2$ -type spectrum in the idealized continuous limit.

Thus accumulation alone does not produce canonical 1/f noise either.

## 4.3 1/f noise requires scale-free hierarchy

Canonical 1/f noise requires an additional structural condition: unresolved processes or integration depths must appear across many timescales with no privileged scale. In practice this often means an approximately log-uniform distribution of relaxation or integration times.

Therefore the correct structural claim is:

1/f noise appears when discrete observation or reconstruction crosses a scale-free hierarchy of unresolved timescales. It does not follow from discreteness alone.

This distinction preserves the formal content of the framework and prevents the overclaim that any sampling of a continuum must generate pink noise.

## 5. The Superposition Derivation of 1/f Noise

The most transparent derivation starts with relaxation processes.

A single relaxation process with characteristic time  $\tau > 0$  contributes a Lorentzian spectrum

$$S_\tau(f) = \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2},$$

where  $\sigma^2$  is a variance scale.

Suppose the observed process aggregates many independent or weakly coupled modes with relaxation times  $\tau$  distributed according to a density  $g(\tau)$  on

$$[\tau_{\min}, \tau_{\max}].$$

The aggregate spectrum is

$$S(f) = \int_{\tau_{\min}}^{\tau_{\max}} \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2} g(\tau) d\tau.$$

Assume the system has no preferred scale across this range, so that the density is uniform in logarithmic scale:

$$g(\tau) d\tau = c \frac{d\tau}{\tau}.$$

Equivalently,

$$g(\tau) = \frac{c}{\tau}.$$

Substitution gives

$$S(f) = 2\sigma^2 c \int_{\tau_{\min}}^{\tau_{\max}} \frac{1}{1 + (2\pi f\tau)^2} d\tau.$$

Let

$$u = 2\pi f\tau,$$

so that

$$d\tau = \frac{du}{2\pi f}.$$

Then

$$S(f) = \frac{\sigma^2 c}{\pi f} [\arctan(u)]_{2\pi f\tau_{\min}}^{2\pi f\tau_{\max}}.$$

In the intermediate frequency regime

$$\frac{1}{\tau_{\max}} \ll f \ll \frac{1}{\tau_{\min}},$$

we have

$$2\pi f\tau_{\min} \ll 1, \quad 2\pi f\tau_{\max} \gg 1.$$

Therefore

$$\arctan(2\pi f\tau_{\max}) \approx \frac{\pi}{2}, \quad \arctan(2\pi f\tau_{\min}) \approx 0.$$

Hence

$$S(f) \approx \frac{\sigma^2 c}{\pi f} \cdot \frac{\pi}{2} = \frac{\sigma^2 c}{2f}.$$

Thus

$$S(f) \propto \frac{1}{f}.$$

This is the rigorous mathematical core of the proposed frame-edge account.

## 6. Theorem Schema: Scale-Free Frame-Edge Spectral Residue

**Theorem Schema.** Let a translation

$$T : P_C \rightarrow P_L$$

represent a higher global or scale-rich process inside a lower discrete observational frame. Suppose the effective error, memory, or relaxation content of the translation decomposes into independent or weakly coupled modes with Lorentzian spectra

$$S_\tau(f) = \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2},$$

and suppose the unresolved timescales are approximately log-uniformly distributed on

$$[\tau_{\min}, \tau_{\max}].$$

Then, for frequencies satisfying

$$1/\tau_{\max} \ll f \ll 1/\tau_{\min},$$

the aggregate spectrum satisfies

$$S(f) \sim \frac{C}{f}$$

for a positive constant  $C$  determined by the variance scale and density normalization.

**Interpretation.** Under these hypotheses,  $1/f$  noise is the spectral residue of a scale-free frame transition. It records the fact that the lower frame has not resolved the higher process at one characteristic scale, but across a hierarchy of scales.

## 7. Generalized Exponent Case

The log-uniform case gives canonical  $1/f$  noise. A nearby power-law distribution of timescales gives nearby exponents.

Let

$$g(\tau) = c\tau^{-\gamma}$$

on a wide timescale interval. Then

$$S(f) = 2\sigma^2 c \int \frac{\tau^{1-\gamma}}{1 + (2\pi f\tau)^2} d\tau.$$

With  $u = 2\pi f\tau$ , this becomes, up to multiplicative constants,

$$S(f) \propto f^{\gamma-2} \int \frac{u^{1-\gamma}}{1 + u^2} du.$$

When the intermediate-range integral is approximately frequency independent, one obtains

$$S(f) \propto \frac{1}{f^\alpha}, \quad \alpha = 2 - \gamma.$$

For  $\gamma = 1$ ,

$$\alpha = 1.$$

Thus canonical pink noise appears as the log-uniform case, while neighboring hierarchical distributions produce  $1/f^\alpha$  spectra near  $\alpha = 1$ .

## 8. Probability and $1/f$ as Different Layers of the Same Seam

Probability and  $1/f$  noise should not be collapsed into the same claim.

Probability concerns the relation between sampled outcomes and the measure law under which they are evaluated. This already exhibits a frame distinction:

$$\text{finite sample} \neq \text{global probability law}.$$

$1/f$  noise concerns the spectral structure of fluctuations when sampling, integration, or reconstruction occurs across a hierarchy of unresolved timescales.

Thus:

- Probability supplies the continuum-side law.
- Randomness appears in the sampled path relative to that law.
- $1/f$  noise appears when the sample-law relation becomes mediated by scale-free unresolved memory, relaxation, or integration hierarchy.

In Lambda vocabulary, randomness is a sample-measure residue, while 1/f noise is a spectral hierarchy residue. They are related, but not identical.

## 9. Transcendence as a Separate Typed Residue

The constants  $\pi$  and  $e$  should not be described as beyond arithmetic in every sense. Both are computable and definable by finite formulas, algorithms, limits, series, and integrals. The precise theorem-level statement is narrower:

- $\pi$  and  $e$  are transcendental.
- Therefore neither is a root of any nonzero polynomial with rational coefficients.
- They exceed the algebraic frame over  $\mathbb{Q}$ , not every formal or arithmetic frame.

In Lambda terms, transcendence is algebraic-frame residue:

$$\pi, e \notin \overline{\mathbb{Q}}.$$

This typed residue parallels, but does not equal, probabilistic randomness or 1/f spectral residue. The shared pattern is frame non-collapse: an admissible lower frame attempts to close over a richer target and encounters a structured remainder.

## 10. Nonstandard Sampling and Overspill

Nonstandard models make the sampling seam especially visible. In a hyperfinite model, one can consider sequences of length

$$H \in {}^*\mathbb{N} \setminus \mathbb{N}.$$

Such sequences may agree with all standard finite tests up to every standard bound while still exhibiting global deviations at nonstandard scales. This does not show that ordinary coin tosses produce 1/f noise. Rather, it shows a different frame-edge fact:

Local finite agreement does not automatically determine global behavior over nonstandard horizons.

This mirrors the general Lambda pattern. Standard finite verification can preserve local behavior while failing to close over global, nonstandard, or continuum-level structure.

A fully formal nonstandard theorem requires specifying:

1. the model;
2. the internal sample sequence;
3. the class of standard tests preserved;
4. the global property that fails or deviates;
5. the transfer, compactness, or overspill principle used.

With those declarations, nonstandard sampling supplies a rigorous model of frame-edge residue.



## 11. Relation to RH and Prime Distribution

The analogy with the Riemann Hypothesis should be stated at the level of frame structure, not ordinary stochastic identity.

In RH, the lower frame contains arithmetic traces: primes, von Mangoldt sums, Mertens sums, divisor functions, and other discrete number-theoretic data. The higher frame contains analytic-spectral structure:  $\zeta(s)$ , analytic continuation, the functional equation, the completed  $\xi$ -function, zero distribution, and critical-line rigidity.

The analogy is:

$$\text{discrete samples / arithmetic traces} \leftrightarrow \text{global measure / analytic-spectral law.}$$

RH is not a claim about coin tosses. It is a claim about exact analytic structure controlling arithmetic distribution. But the same Lambda architecture appears: the lower discrete trace does not by itself close over the higher global law without a bridge. The bridge carries residue unless the relevant analytic-spectral strength has been supplied.

## 12. Ontological Claim, Precisely Stated

The supported ontological statement is:

A lower discrete observational frame and a higher global measure or spectral frame do not describe the same object in the same way. A finite sample is not the measure. An empirical frequency is not the probability law. A raw independent sequence is not a  $1/f$  process. A  $1/f$  spectrum appears when the frame transition contains a scale-free hierarchy of unresolved integration or relaxation timescales. In that case, the spectrum records a real structural feature of the translation seam.

This gives the Lambda / TuFT / NSAF reading a precise foundation. It does not require mystical randomness, vague emergence, or unqualified inevitability. It requires declared frames, declared laws, and declared bridge mechanisms.

## 13. Final Synthesis

The central synthesis can now be stated as follows.

Randomness and probability reveal a basic frame distinction: discrete outcomes occur in the linear frame, while probability laws live in the global measure frame. Repeated sampling can estimate, test, and approximate the law, but no finite string is identical with the measure object.

$1/f$  noise reveals a more specific spectral condition at such seams. It arises when reconstruction or observation must integrate over unresolved timescales distributed approximately uniformly on a logarithmic scale. The resulting spectrum

$$S(f) \sim \frac{1}{f}$$

is therefore not a universal consequence of sampling. It is the signature of scale-free hierarchical sampling or relaxation.

Within Lambda vocabulary:

- randomness is sample-measure residue;
- transcendence is algebraic-frame residue;
- nonstandard drift is model-theoretic sampling residue;
- $1/f$  noise is spectral hierarchy residue;
- RH-style error control is arithmetic / analytic-spectral translation residue.

These are not identical phenomena, but they share a common architecture: a declared lower frame attempts to stabilize a richer target, and the structured remainder records the boundary of that frame.

This is the original assertion:  $1/f$  noise becomes mathematically and ontologically intelligible as the spectral signature of a scale-free frame edge, while randomness and probability provide the most accessible case of the deeper sampling-law distinction.